



MODELING IN INFECTIOUS DISEASES

MODELING TEAM - HCDC

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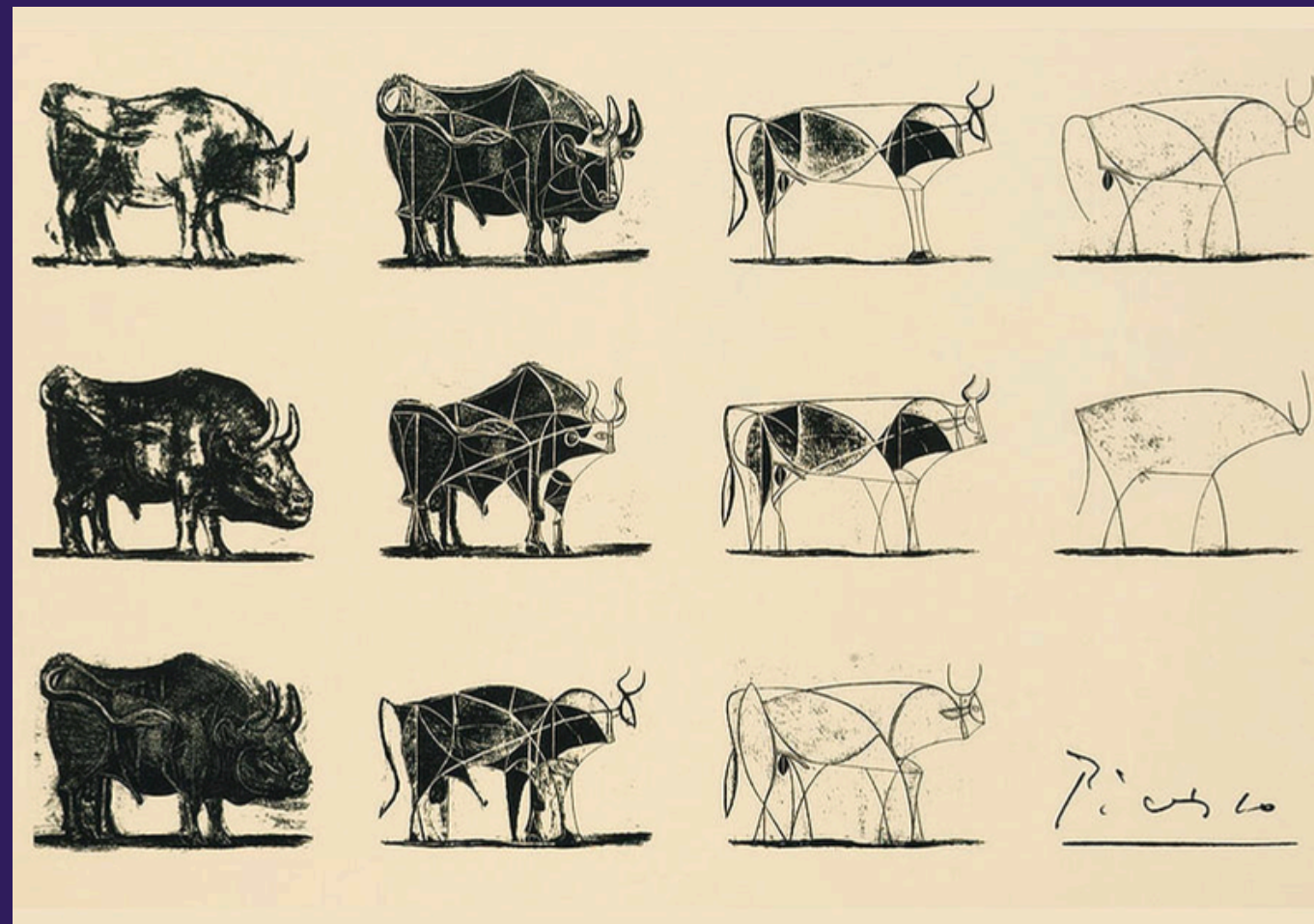
04

Agent Based Model



WHAT IS A MODEL?

What is a Model?



The Bull, by Pablo Picasso

“A model is a simplified representation of a complex phenomenon” (Vynnycky & White, 2010)

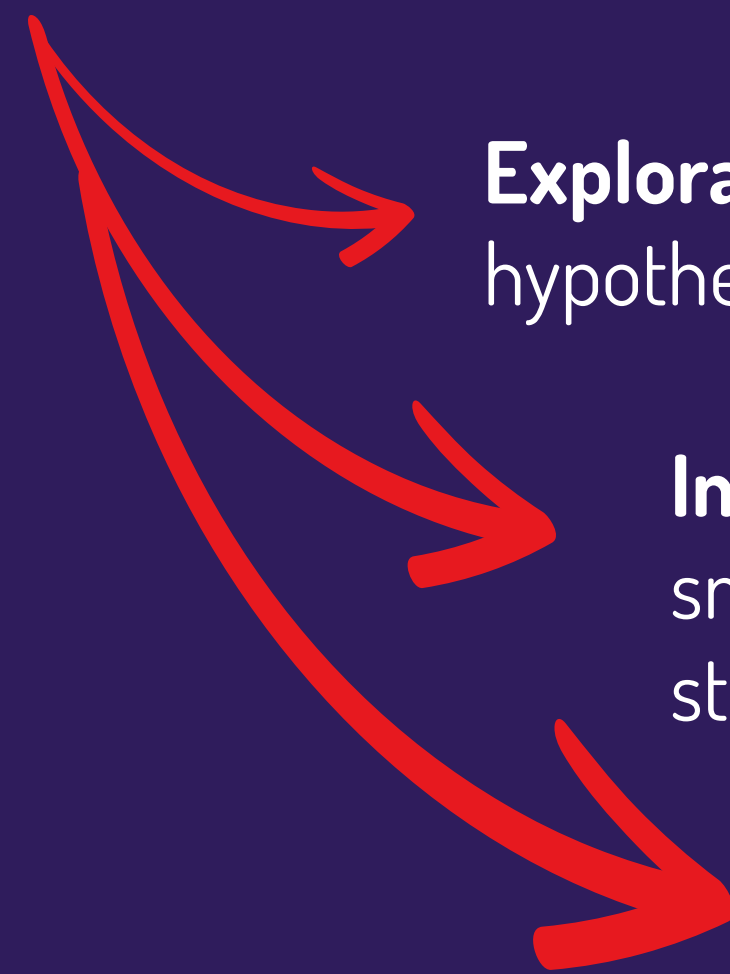
Models help us **navigate** and **make decisions** about complex situations by **focusing on the most relevant factors** and **ignoring the less critical details**.

- The transmission of infection is a stochastic process, such that no two epidemics are identical.
- Models will always be an approximation, and rare or unforeseen behavioral events can have a significant impact on the disease dynamics.

There is no perfect model

What is a Model?

Purposes of modelling



Exploration: to identify patterns in the data and generate hypotheses (then we use inference to test these hypotheses).

Inference: estimate parameters or conduct hypothesis testing, using a small fraction sampling from the population, and try to draw some statements about the whole population.

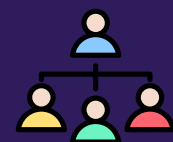
Prediction: the key goal is to find a model that can accurately predict y for new observations.

What is a Model?

Deterministic models

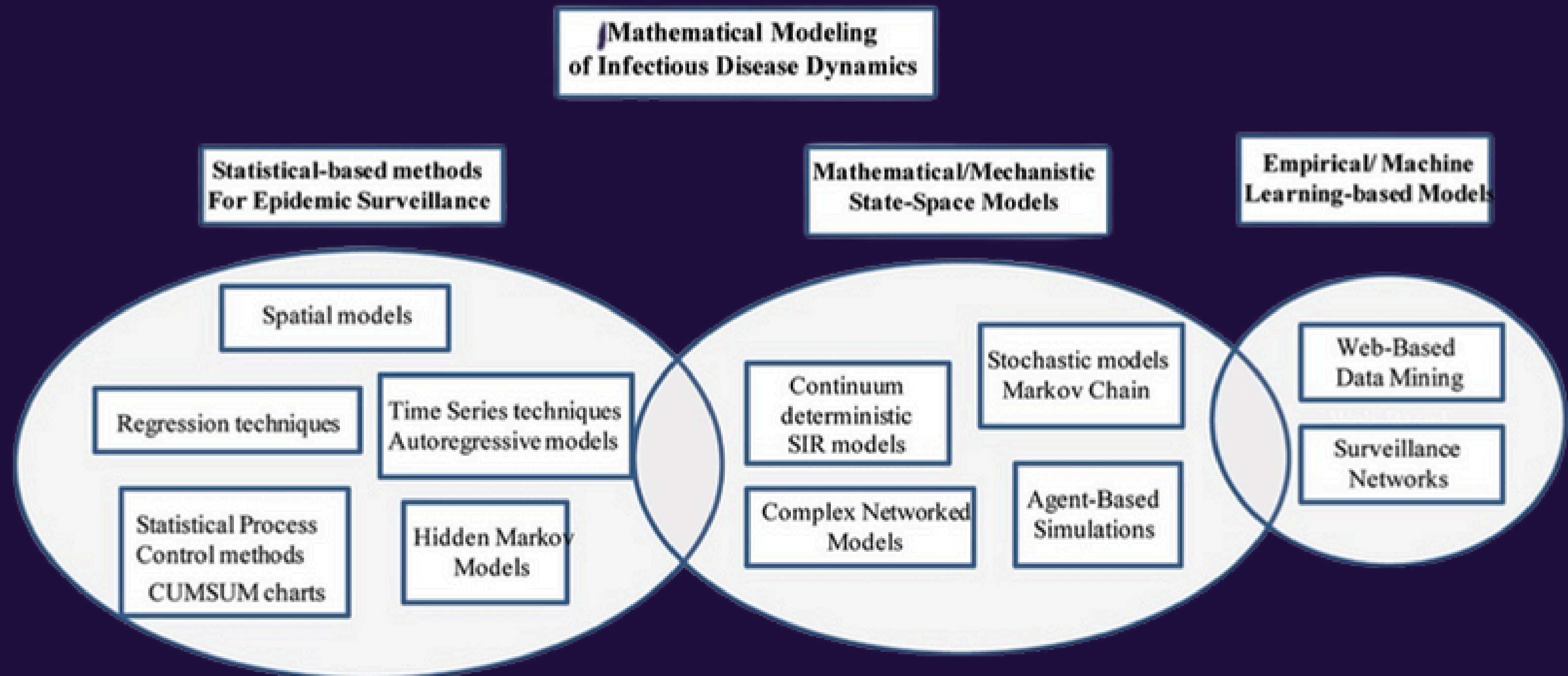


- **Time dynamic:** dynamic or static
- **Randomness:** stochastic / probabilistic or deterministic
- **Behavior:** population or individual
- **Time representation:** discrete or continuous
- **Entry:** open or closed
- **Equation structure:** linear or nonlinear

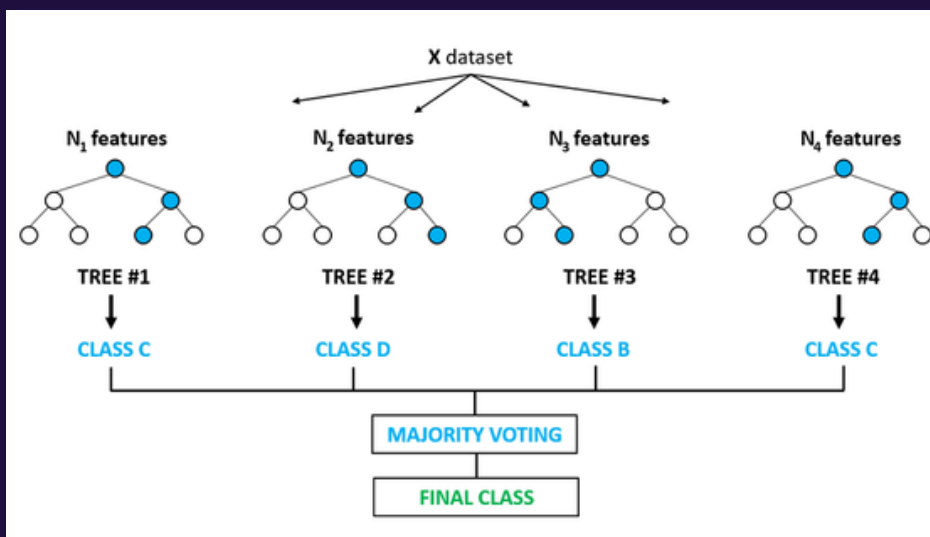
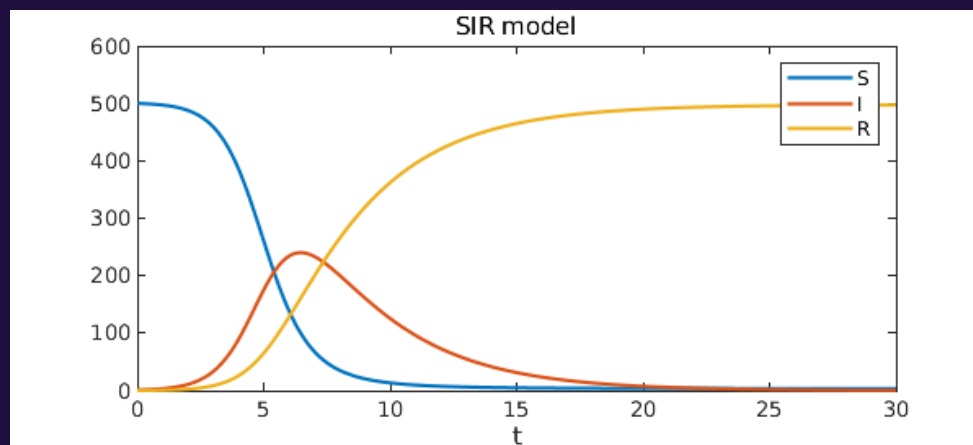
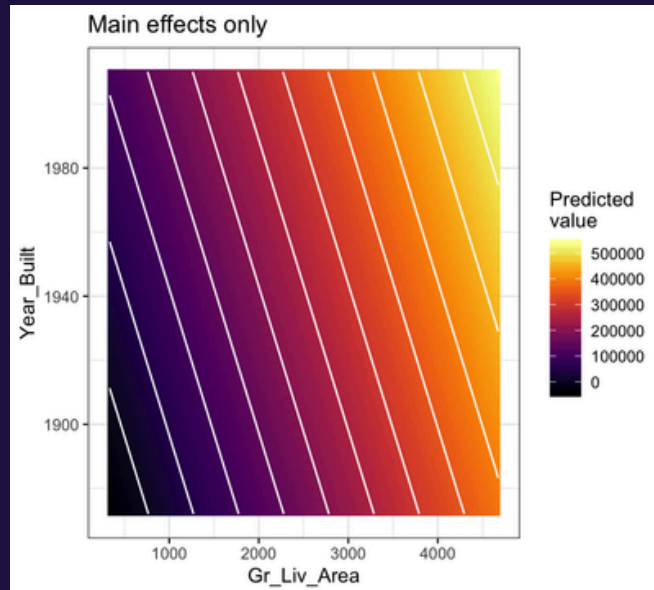


Stochastic models

Model Classification



Model Classification



Statistic Model

Dựa vào xác suất và phân phối dữ liệu để xác định mối quan hệ giữa các biến số



Giám sát dịch bệnh, dự đoán xu hướng dịch tễ từ dữ liệu thực tế.

Mechanistic Model

Sử dụng phương trình toán học để mô tả sự tiến triển của hệ thống theo thời gian



Mô phỏng sự lây lan của dịch bệnh dựa trên các giả định

Machine Learning Model

Sử dụng các thuật toán máy học, ý kiến chuyên gia để phân tích dữ liệu và dự đoán diễn biến dịch bệnh



Xử lý dữ liệu lớn, nhận diện các xu hướng phức tạp

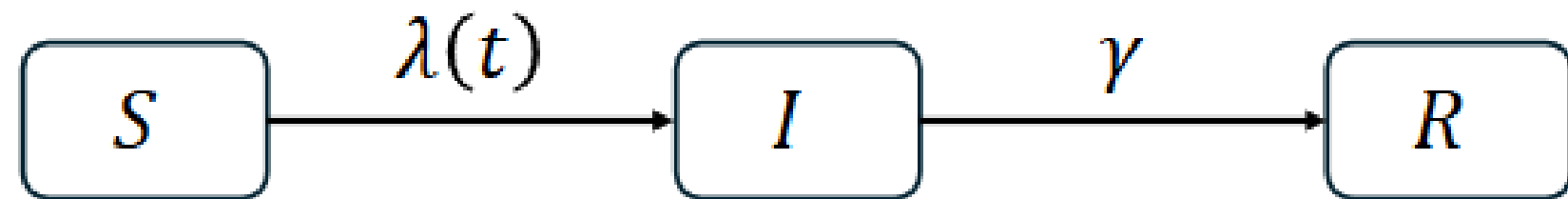


DETERMINISTIC AND STOCHASTIC MODELS

Deterministic and Stochastic models

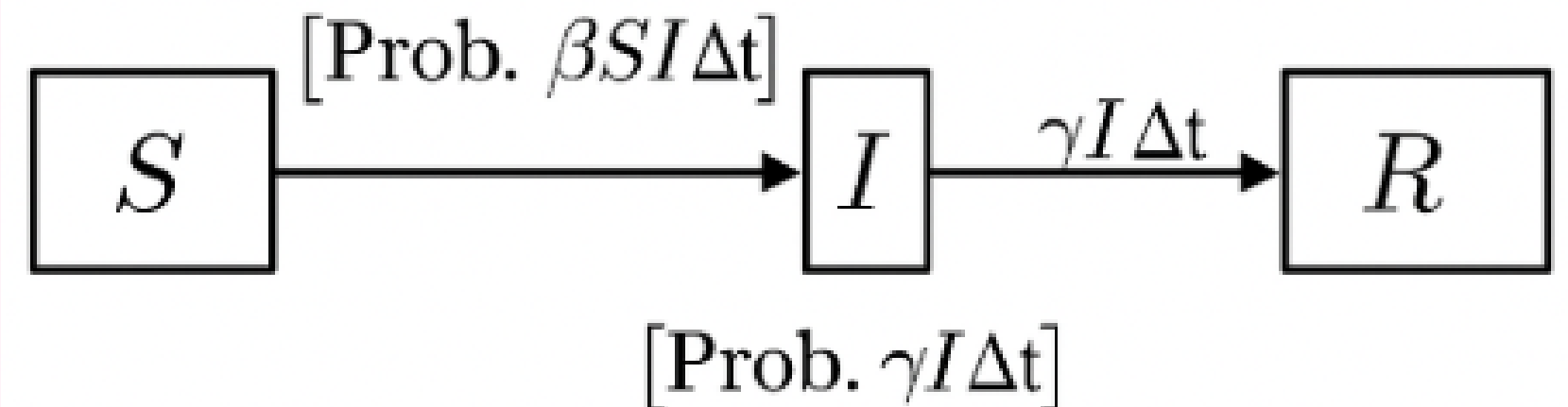
Deterministic models

Assumptions that **there is no randomness**, and the system can be fully described by a defined set of equations and parameters. **Each run** of model always generates the **same result**



Stochastic models

Assumptions that events occur **randomly**, affecting single parameters or multiple components within the system. **Each run** of a model generates **different results**.

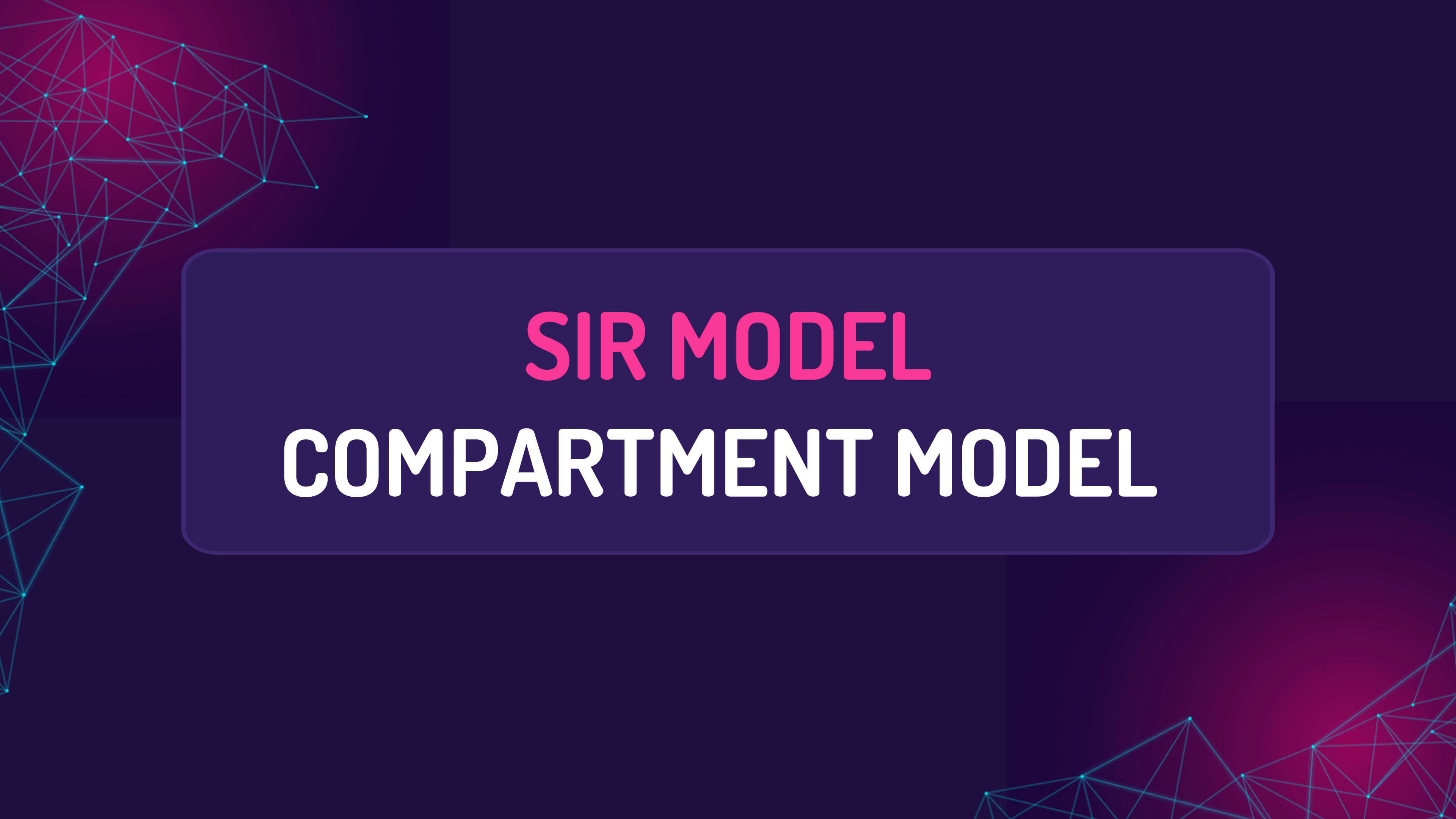


Therefore, stochastic models must be run for hundreds or thousands of iterations to explore the full range of possible outcomes



Deterministic and Stochastic models

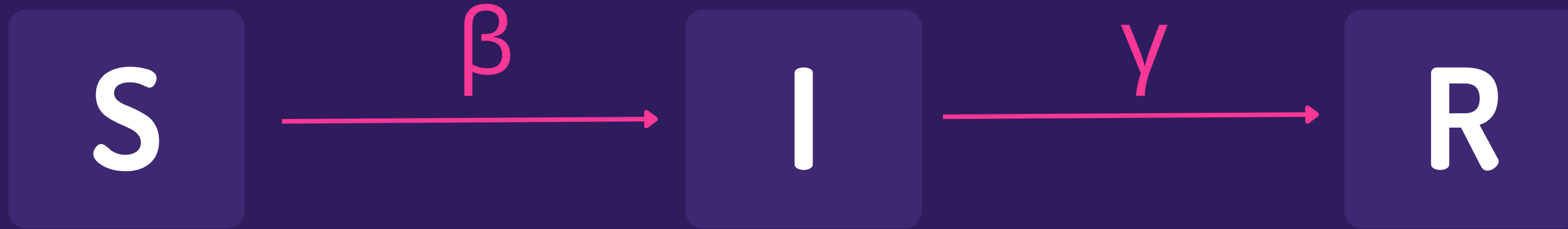
	Deterministic	Stochastic
Randomness	No	Yes
Output	The same	Different
Implement	Easy	Complex
Calculate	Fast	Simulation, repeat
Probability	No	Yes



SIR MODEL

COMPARTMENT MODEL

SIR model



$$dS/dt = -\beta SI$$

$$dI/dt = \beta SI - \gamma I$$

$$dR/dt = \gamma I$$

- Parameters: β (transmission rate); γ (recovery rate)
- The negative sign means the number of susceptible people decreases over time as they get infected.
- $N = S + I + R$
- $dS/dt + dI/dt + dR/dt = dN/dt = 0$ describes a closed population
- In mathematical models, N is ideally constant, but in reality, N can change.

SIR model



Kermack-McKendrick model Susceptible-Infectious-Recovered (SIR) model

$$dS/dt = -\beta SI$$

$$dI/dt = \beta SI - \gamma I$$

$$dR/dt = \gamma I$$

Assumptions:

- Closed population;
- Density-dependent transmission;
- Instantaneous incubation period;
- Duration of infectivity equals disease length;
- Homogeneous population;
- No reinfection

FREQUENCY-DEPENDENT TRANSMISSION AND DENSITY-DEPENDENT TRANSMISSION

Frequency-Dependent

Depends on the frequency of contact between individuals

The number of contacts/individuals remains **constant** regardless of population size

Common in stable contact network (HIV/AIDS - personal behaviors)

$$\frac{dS}{dt} = -\beta S \frac{I}{N} \quad \text{Unit of } \beta: \frac{1}{\text{time}}$$

Easier control (vaccination reduces effectively)

Density-Dependent

Depends on the density of the population

The number of contacts/individual **increases** with population density

Common in general population interactions (Influenza - respiratory droplets)

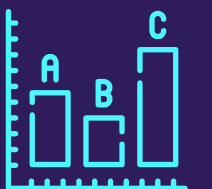
$$\frac{dS}{dt} = -\beta SI \quad \text{Unit of } \beta: \frac{1}{\text{time} \times \text{people}}$$

Harder control (reducing crowding or increasing physical distancing)

DETERMINING R_0

BASIC THEORY

- **Definition:** the average number of secondary infections from one infected person in a fully susceptible population at the start of an epidemic.
- **Significance:** helps determine whether an infectious disease can spread
- **Threshold Value:**
 - If ($R_0 > 1$): infection will spread;
 - If ($R_0 < 1$): infection will die out.
- **Influence Factors:** contact rate, transmission probability/contact, infectious period duration
- **Estimation:** using mathematical models with simulations or assumptions about disease transmission dynamics.
- **Interventions:** vaccination, social distancing can reduce R_0 by altering transmission dynamics.



DETERMINING R0

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Consider the early stage of the epidemics, where $S \approx N$
At equilibrium or in the early phase, $dI/dt \approx 0$
the rate of change of I:

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I \approx 0$$

Divide through by I
($I \ll N$)

$$\beta \frac{S}{N} - \gamma \approx 0$$

Since $S + I + R = N$ and $S \approx N$ initially,
we have $S/N \approx 1$;
Therefore: $\beta \approx \gamma$

$$\beta \frac{S}{N} \approx \gamma$$

COMPARTMENTAL MODEL

The disease will spread if the number of new infections **exceeds** the number of recoveries, which happens when $\beta > \gamma$,
Divide by γ : $\beta/\gamma > 1$

This ratio reflects the number of new infections caused by a single infectious individual before they recover

=> **definition of R0**

So:
$$R_0 = \frac{\beta}{\gamma}$$

R0 is determined by the parameters β and γ

Serves as a threshold for epidemic spread and also facilitates simulation and prediction in mathematical modelling

Lowering R0 through interventions and comparing R0 across different diseases or settings

I MAX

I MAX



The peak value of the infection rate or the highest number of individuals infected at the peak of an epidemic
It helps estimate the healthcare resources needed, such as hospital beds and medical staff, etc. and emphasizes the importance of preventive measures.

$$\frac{dS}{dt} = -\beta \frac{SI}{N}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

To find I **max**, the condition is at the peak of the infection, the rate of change of infectious individuals is zero

$$\frac{dI}{dt} = 0 = \beta \frac{SI}{N} - \gamma I$$

$$\Leftrightarrow S = \frac{\gamma N}{\beta} = \frac{N}{R_0}$$

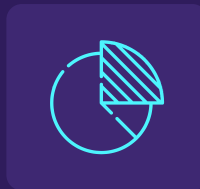
$$\text{At the peak: } I_{max} = N - S_{min} - R_{max}$$

$$\text{Thus: } I_{max} = N \left(1 - \frac{1}{R_0} \right)$$

Measles Outbreak: $N = 10,000$; $R_0 \sim 15 \Rightarrow I_{max} = 9,333$

HERD IMMUNITY THRESHOLD

HERD IMMUNITY



When a significant portion of a population becomes immune to an infectious disease, either through infection or vaccination

To achieve herd immunity, enough individuals need to be immune so that $R_0 < 1$

$$\text{Herd immunity threshold} = 1 - \frac{1}{R_0}$$

$$\frac{dI}{dt} = \beta SI - \gamma I \quad \begin{aligned} N &= S + I + R \\ S &= N - I \end{aligned}$$

quadratic equation

$$\frac{dI}{dt} = \beta NI - \beta I^2 - \gamma I$$

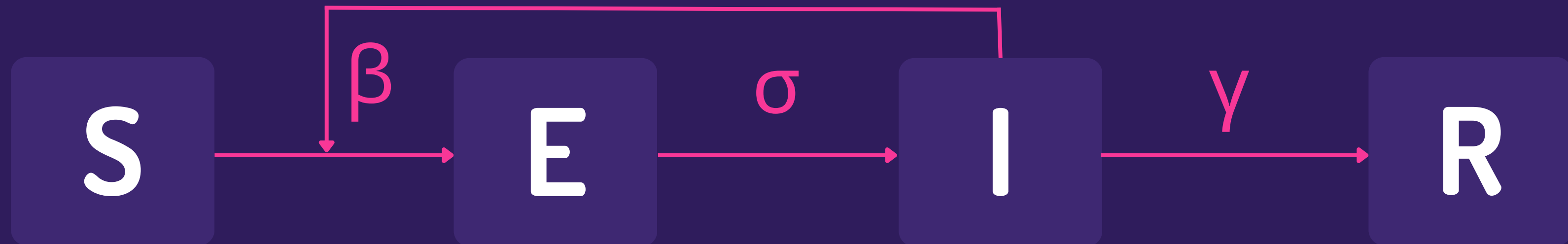
equilibrium points, $y = I$

$$0 = -\beta y^2 + (\beta N - \gamma)y$$

$$y = 0 \quad \text{or} \quad y = \frac{\beta N - \gamma}{\beta} = N - \frac{\gamma}{\beta}$$

$$y = N - \frac{1}{R_0}$$

SEIR model



$$dS/dt = -\beta SI$$

$$dE/dt = \beta SI - \sigma E$$

$$dI/dt = \sigma E - \gamma I$$

$$dR/dt = \gamma I$$

- **Exposed (E):** Individuals who have been infected but are not yet infectious, E accounting for the incubation period.
- Provides more accurate predictions of disease spread and duration
- Aids in public health planning and intervention strategies.

SENSITIVE ANALYSIS AND PARAMETER FITTING

“Odin” package:
creating, simulating,
and analyzing
compartmental
models of infectious
diseases.

Features:

- Model Specification
- Parameter Fitting
- Sensitivity Analysis
- Scenario Analysis
- Integration with Data
- Complex Model Structures

```
library(odin)

# Define the SIR model using odin
sir_model <- odin::odin({
  # Initial conditions
  initial(S) <- 999
  initial(I) <- 1
  initial(R) <- 0

  # Parameters
  beta <- 0.3
  gamma <- 0.1

  # The ODEs
  deriv(S) <- -beta * S * I / N
  deriv(I) <- beta * S * I / N - gamma * I
  deriv(R) <- gamma * I

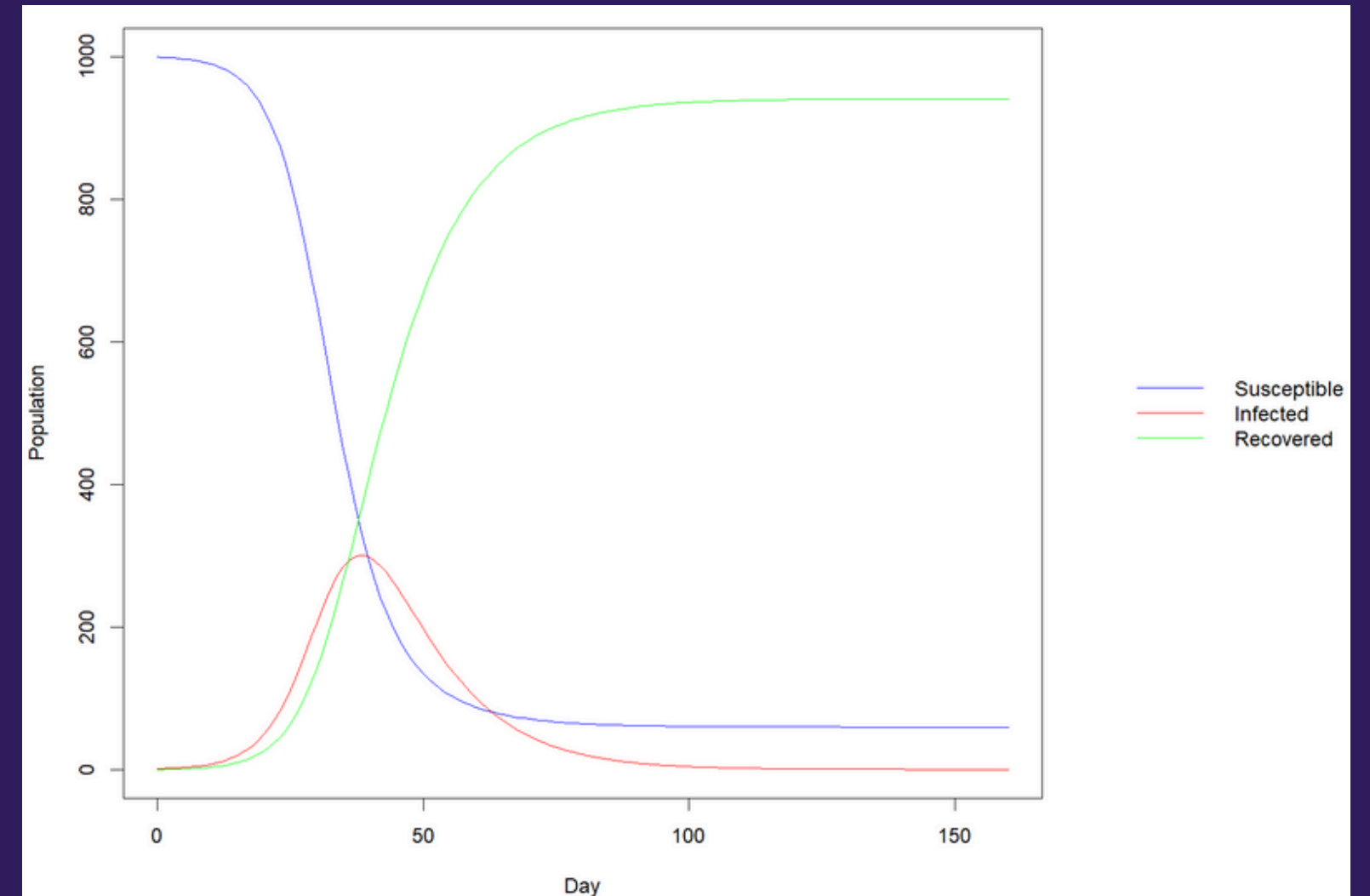
  # Total population
  N <- S + I + R
})

# Generate the model
sir <- sir_model()

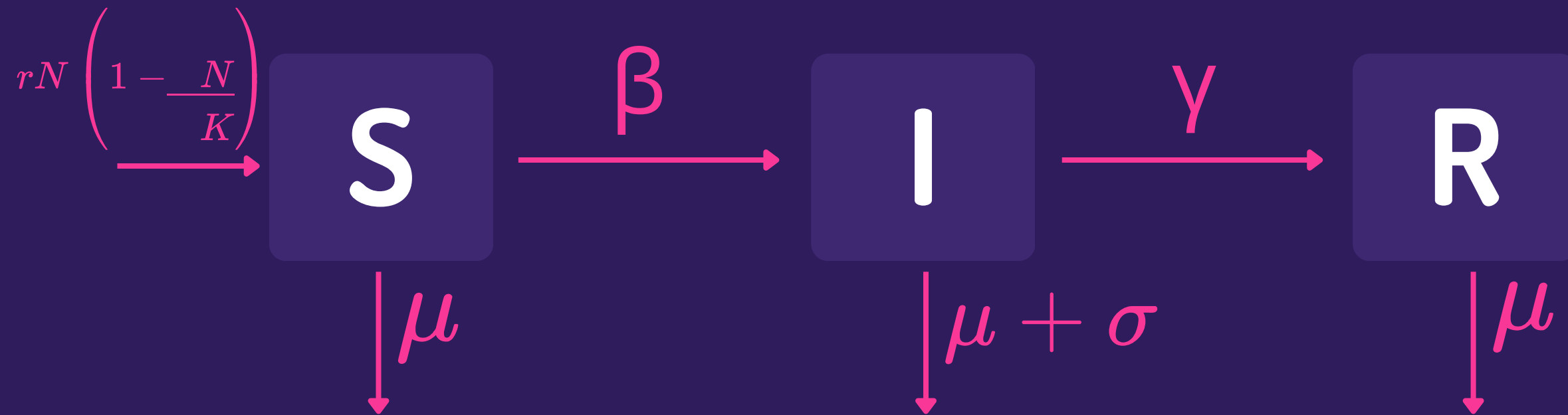
# Time points to solve the model at
times <- seq(0, 160, by = 1)

# Initial state
yini <- c(S = 999, I = 1, R = 0)

# Solve the model
result <- sir$run(times, yini)
```



DYNAMIC POPULATION



$$\frac{dS}{dt} = rN \left(1 - \frac{N}{K}\right) - \beta S \frac{I}{N} - \mu S$$

$$\frac{dI}{dt} = \beta S \frac{I}{N} - \gamma I - I(\mu + \sigma)$$

$$\frac{dR}{dt} = \gamma I - \mu R$$

Parameters

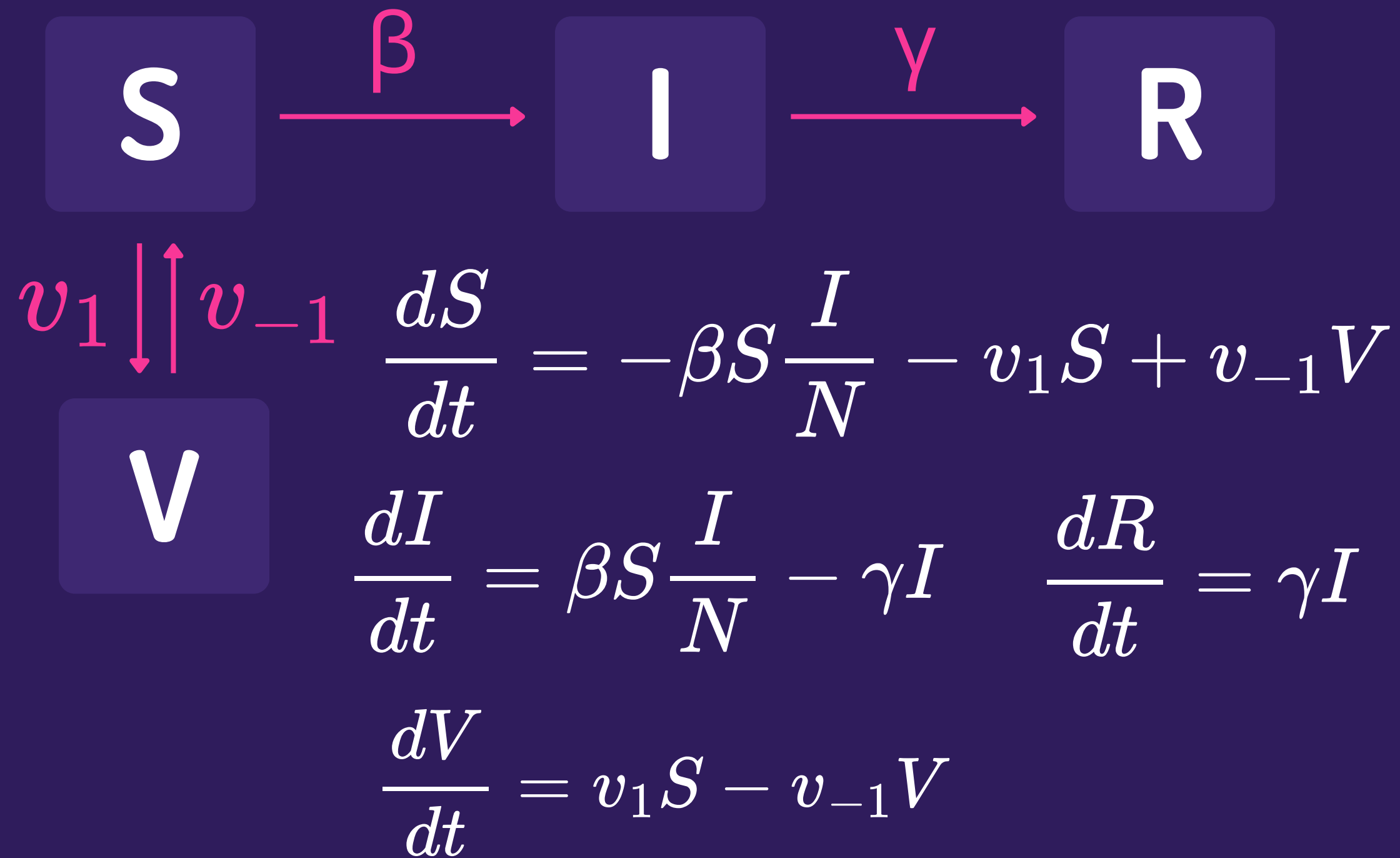
r: per capita rate of increase

K: carrying capacity

μ: natural death rate

σ: disease-related death rate

REPRESENT INTERVENTIONS



VACCINATION

Compartment

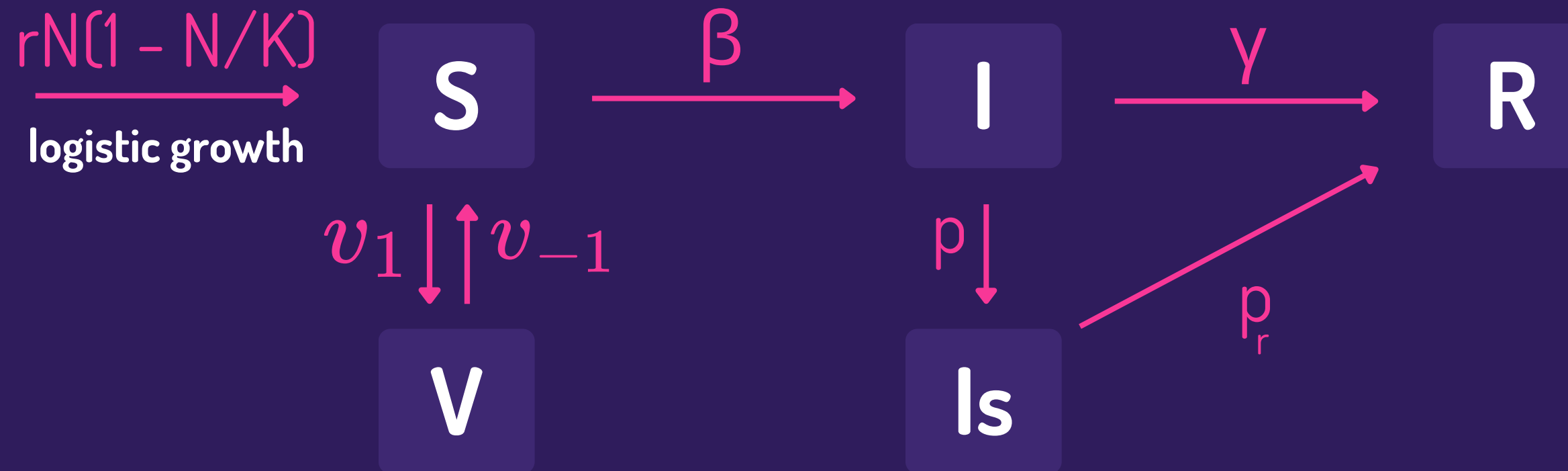
V: vaccinated

Parameters

v1: vaccination rate into the vaccinated compartment

v-1: vaccination rate out of the vaccinated compartment

VACCINATION + ISOLATION



Compartments

V: vaccinated

I_s: isolated

Parameters

p: isolation rate

p_r: isolation to recovery rate

K: carrying capacity

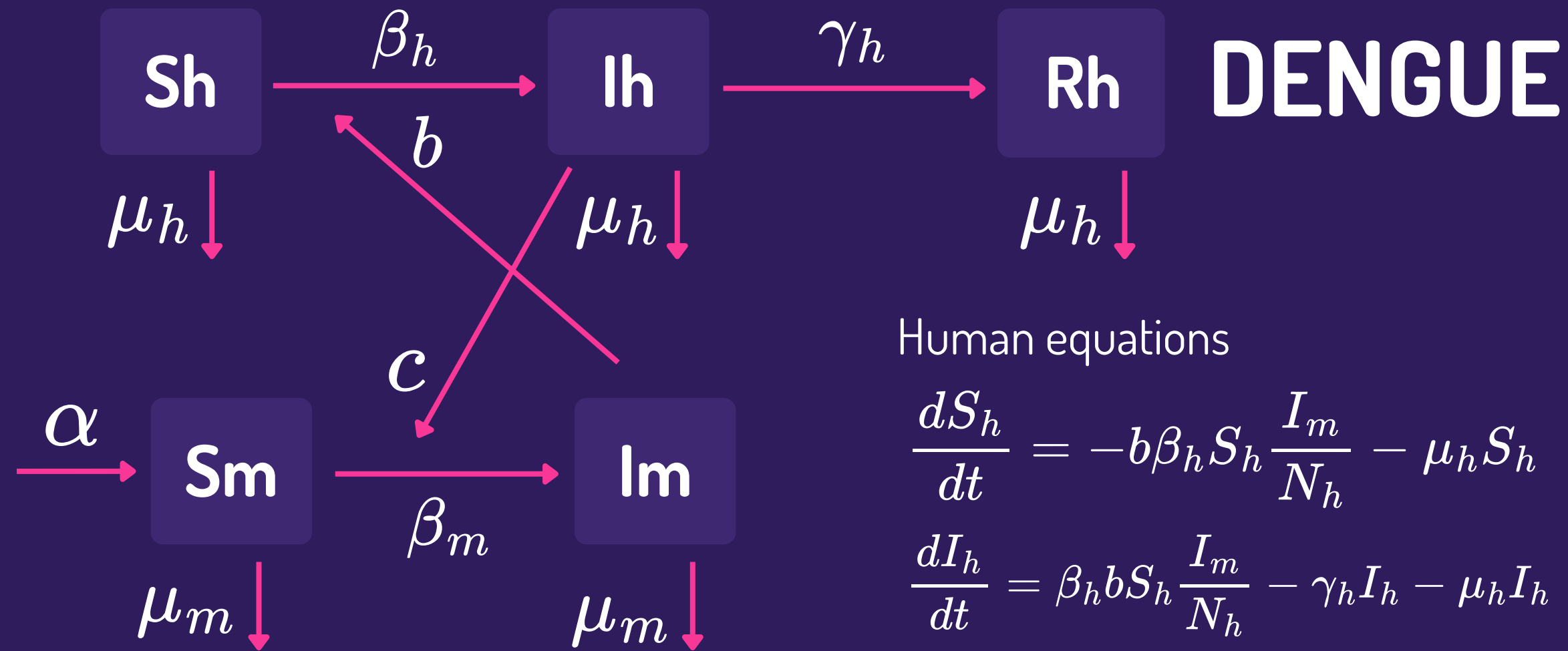
r: intrinsic growth rate

$$\frac{dS}{dt} = rN \left(1 - \frac{N}{K} \right) - \beta S \frac{I}{N} - v_1 S + v_{-1} V$$

$$\frac{dI}{dt} = \beta S \frac{I}{N} - \gamma I - p I_s \quad \frac{dI_s}{dt} = p I - p_r I_s$$

$$\frac{dV}{dt} = v_1 S - v_{-1} V \quad \frac{dR}{dt} = \gamma I + p_r I_s$$

VECTOR-BORNE DISEASE



Human equations

$$\frac{dS_h}{dt} = -b\beta_h S_h \frac{I_m}{N_h} - \mu_h S_h$$

$$\frac{dI_h}{dt} = \beta_h b S_h \frac{I_m}{N_h} - \gamma_h I_h - \mu_h I_h$$

$$\frac{dR_h}{dt} = \gamma_h I_h - \mu_h I_h$$

Mosquito equations

$$\frac{dS_m}{dt} = \alpha N_m - c\beta_m S_m \frac{I_h}{N_h} - \mu_m S_m$$

$$\frac{dI_m}{dt} = \beta_m c S_m \frac{I_h}{N_h} - \mu_m I_m$$

2 populations: humans & mosquitoes

b: The number of humans bitten by one mosquito per unit time.

c: The number of mosquito bites one human receives per unit time.

μ: Natural death

α: The birth rate of mosquitoes

Assumptions:

Homogeneous mixing, constant population size

No latent period, no waning immunity

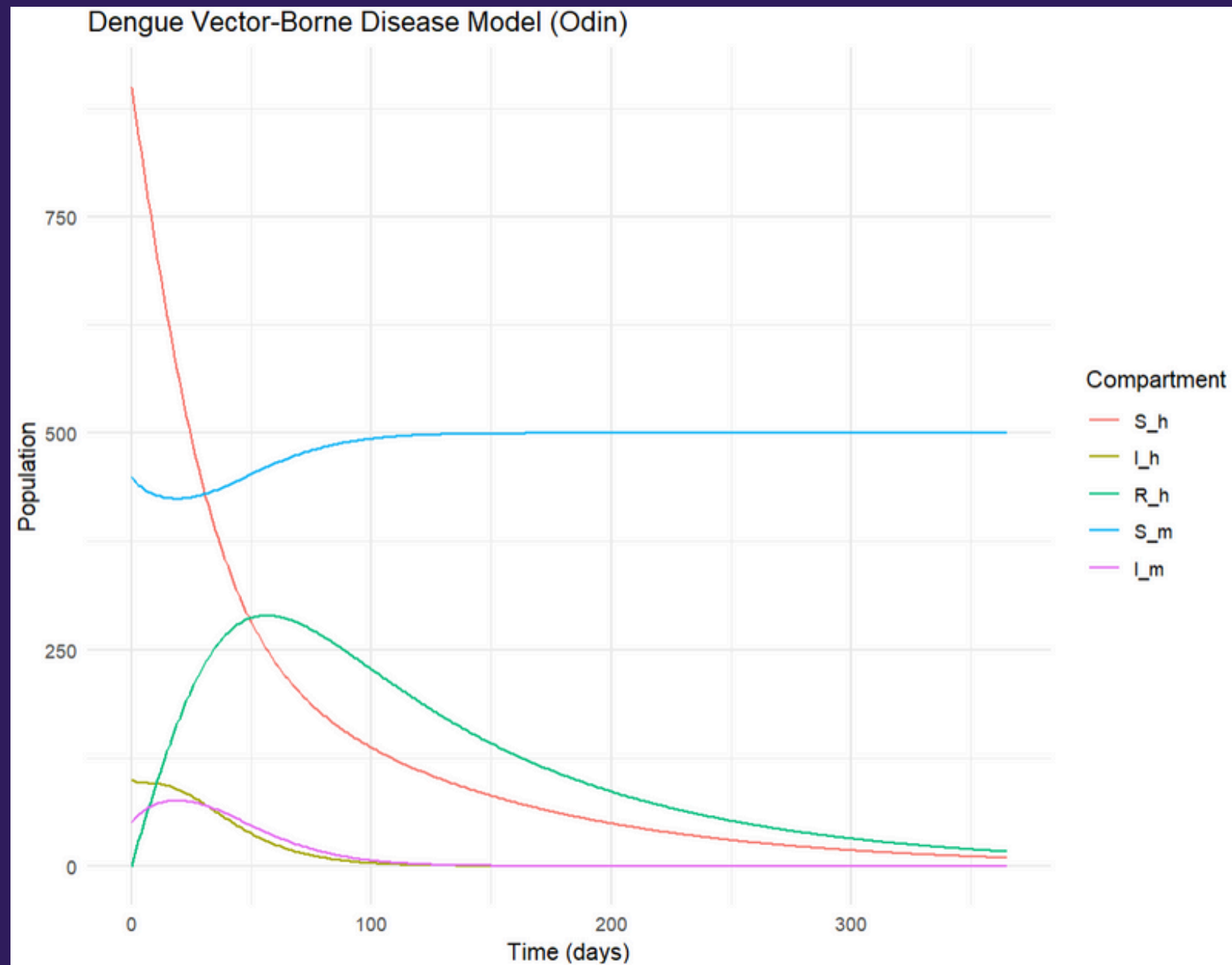
Constant transmission efficiency

Uniform biting rate

Single serotype, no seasonal variations



DENGUE



```
# Define the dengue model using odin
dengue_model <- odin::odin({
  # Parameters
  beta_h <- user(0.2)
  beta_m <- user(0.2)
  b <- user(1)
  c <- user(1)
  gamma_h <- user(0.1)
  mu_h <- user(0.01)
  mu_m <- user(0.1)
  alpha <- user(0.1)
  N_h <- user(1000)
  N_m <- user(500)

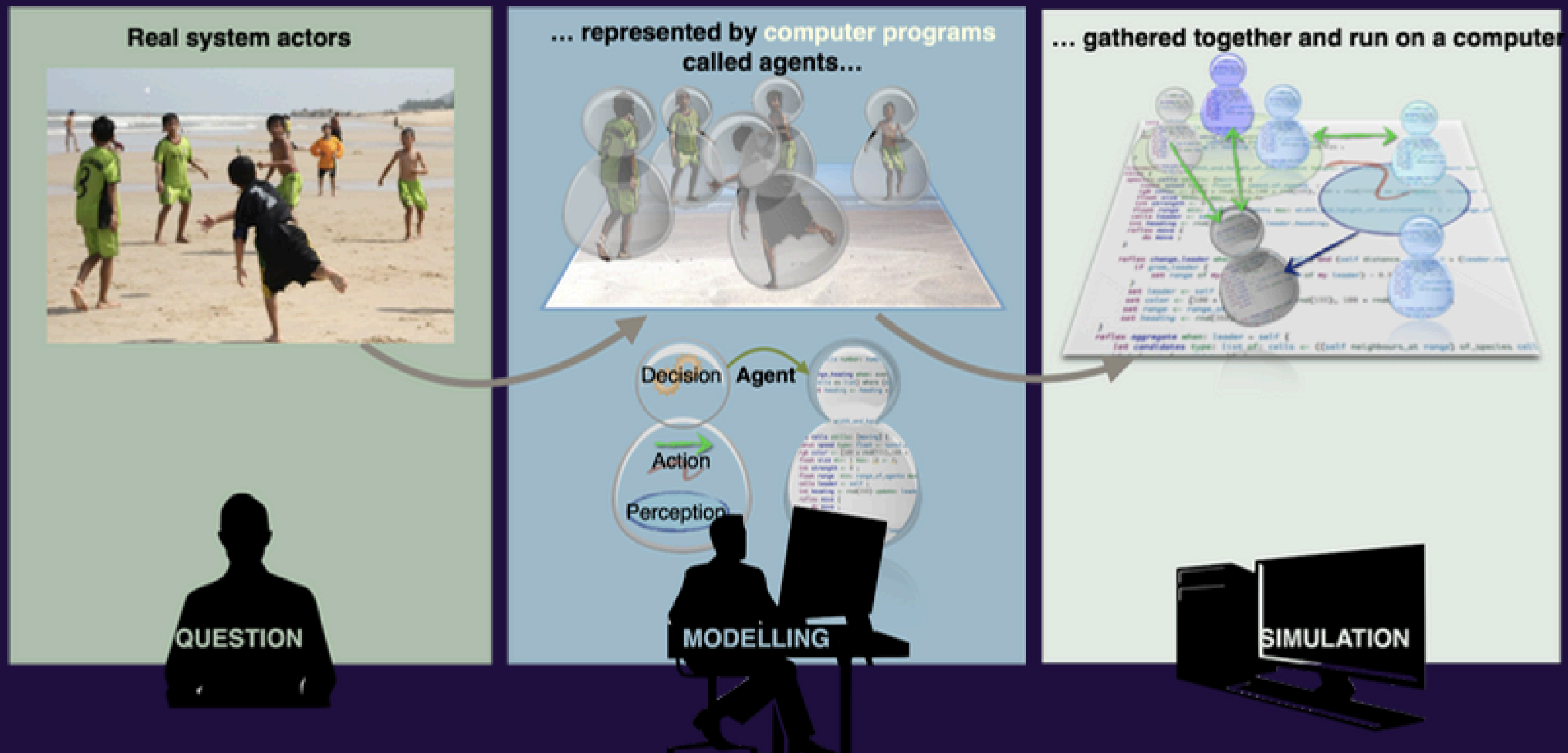
  # Initial conditions
  initial(S_h) <- 900
  initial(I_h) <- 100
  initial(R_h) <- 0
  initial(S_m) <- 450
  initial(I_m) <- 50
})
```



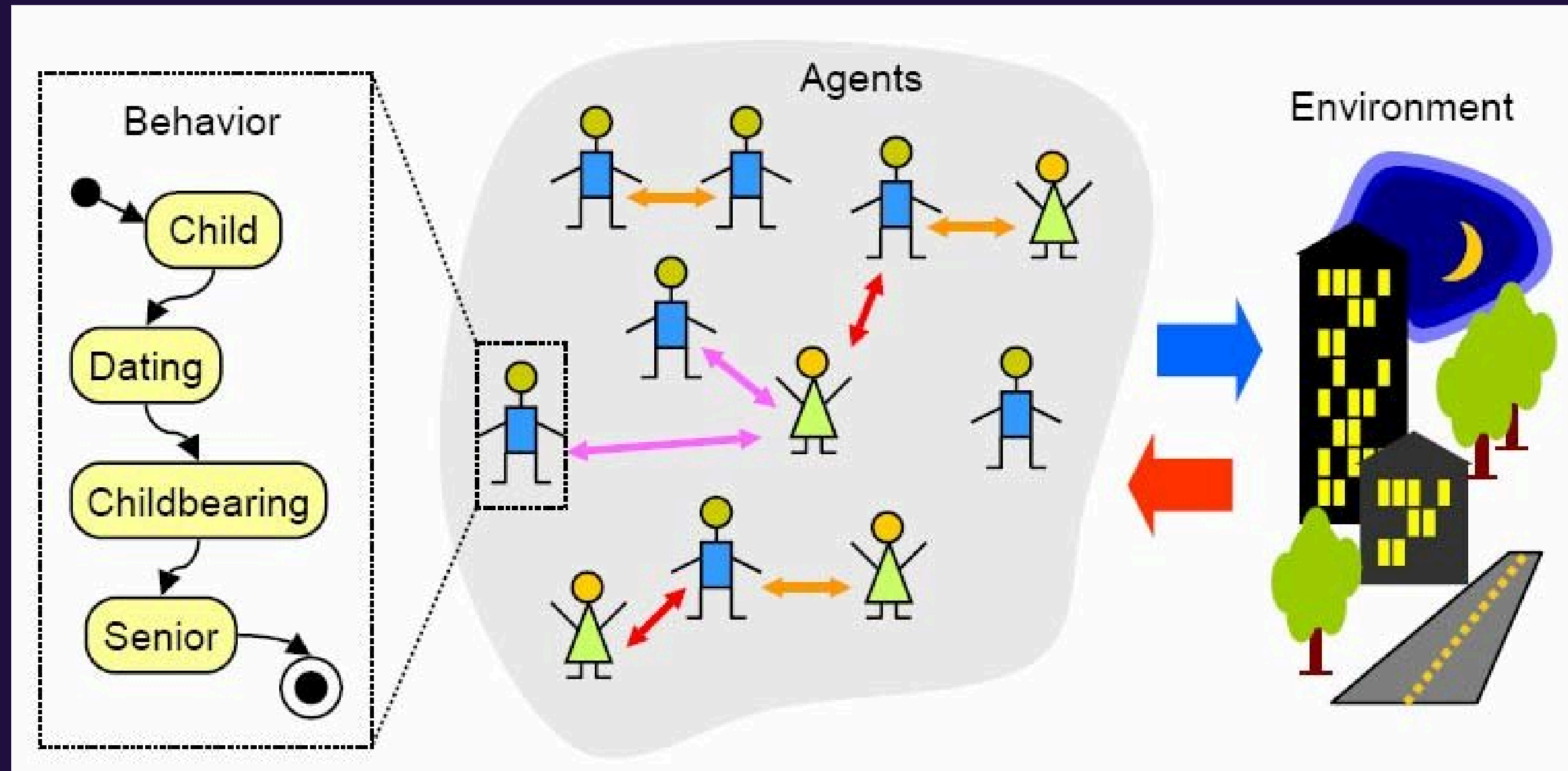
AGENT BASED MODEL

Agent - Based Model

An **individual-centered** representation that allows to reconstruct and simulate **virtual worlds** on a computer.

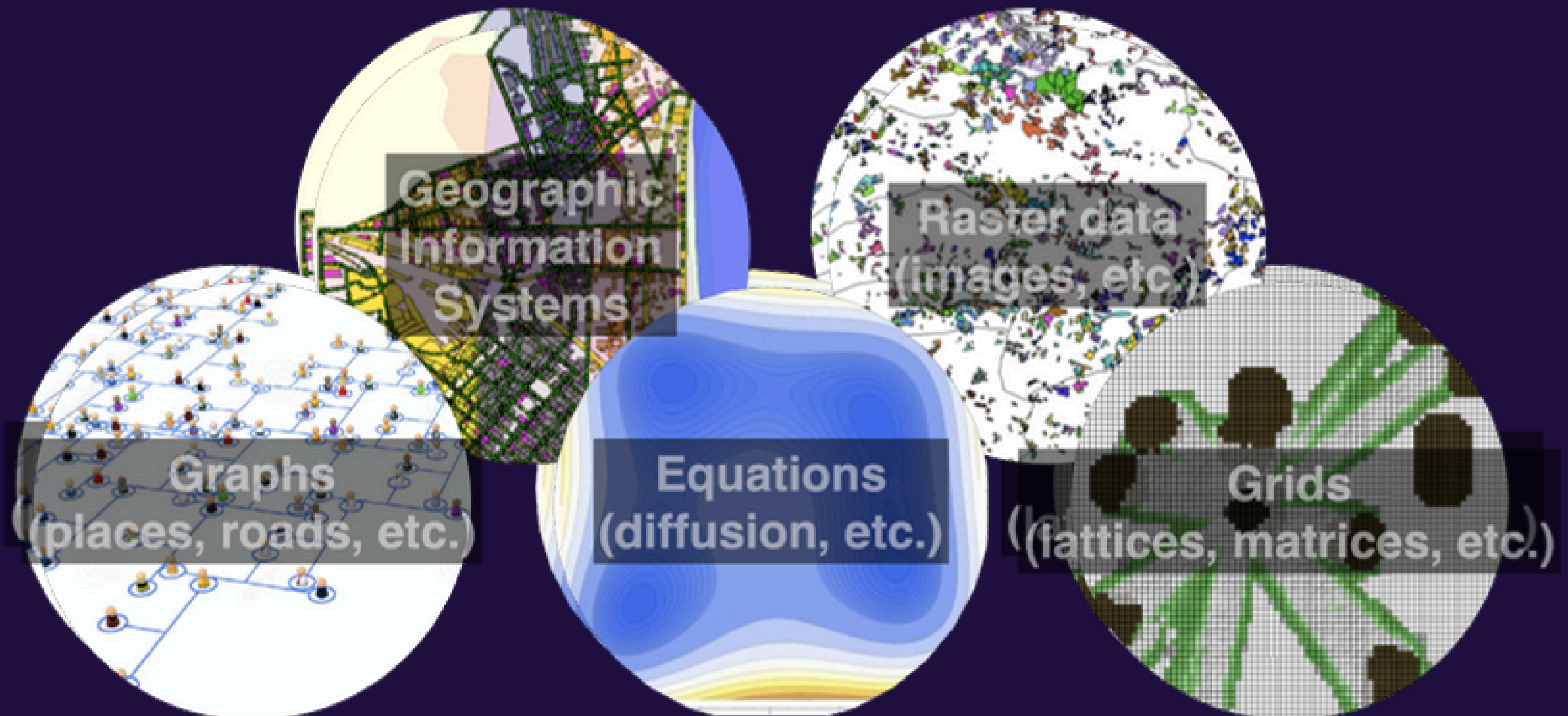


Agent - Based Model



Agent - Based Model

Agent-based models can be spatially explicit: **the environment** can be very detailed and be made up of **spatial agents** itself.

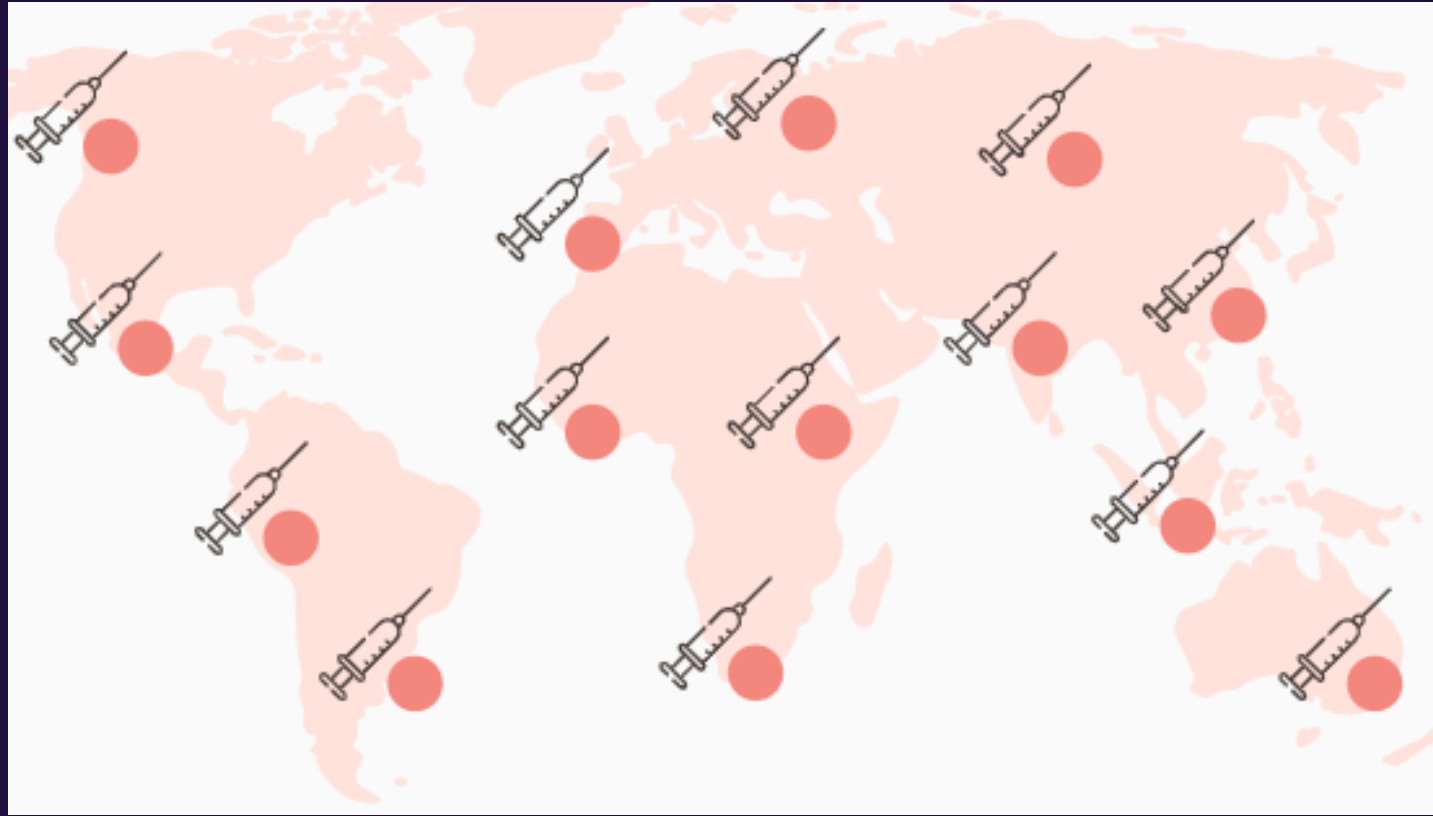


Agent – Based Model

Exploring Regional Outbreak Risks

- How does the risk of outbreaks vary across different areas with different characteristics, including vaccine coverage, mobility, and interactions between citizens?
- How to optimize the vaccine campaign in different areas?

Agent - Based Model



Vaccination Campaign

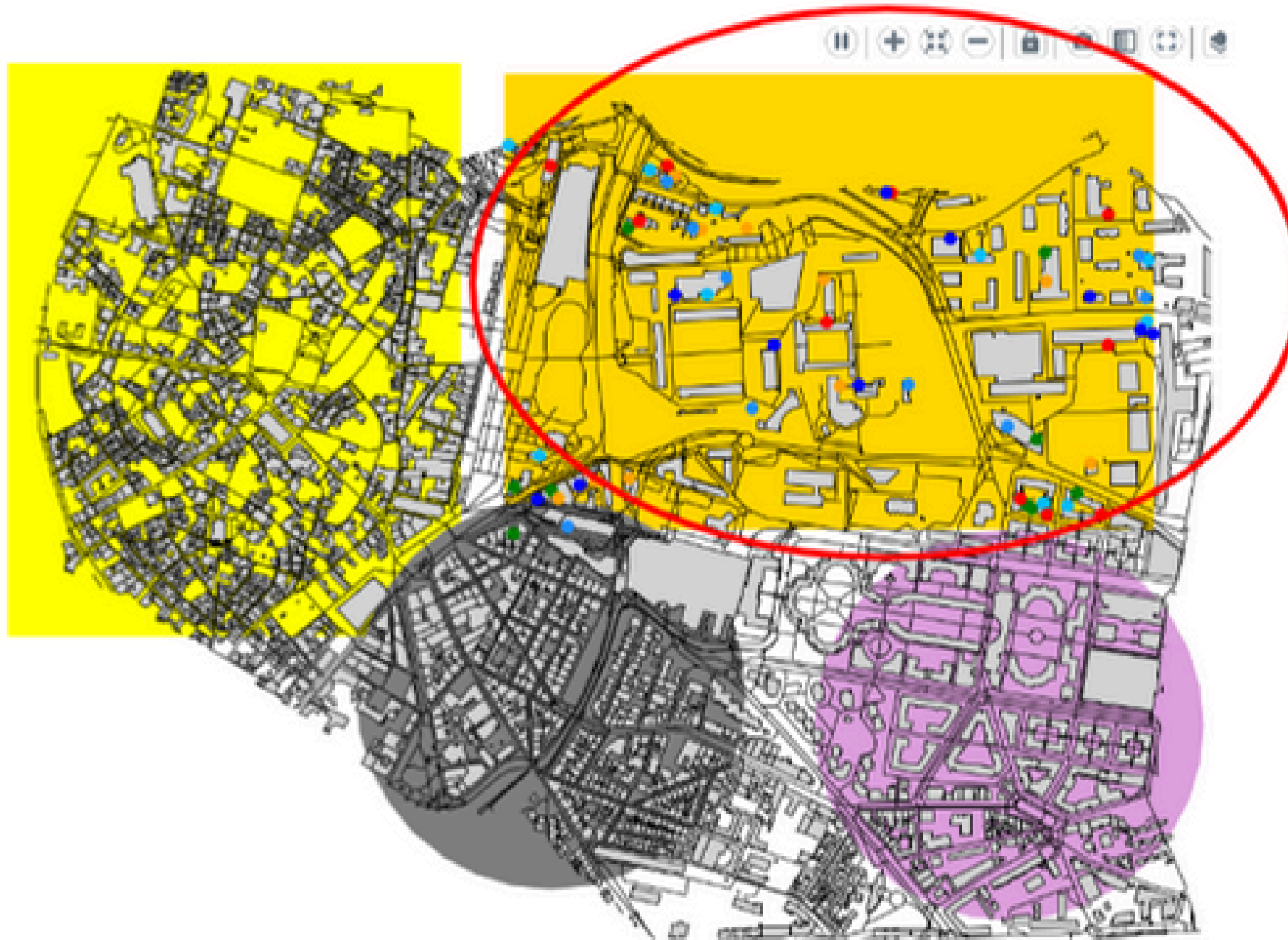


Population characteristics

- **Latent time:** 7 - 18 days
- **Recovery time:** 10 - 12 days
- **Contact distance:** 2 meters
- **Facilities:** 01 hospital, 4 school in each area, 3 playground in A, C, D

Agent - Based Model

Characteristic of **Area A**



States

V1: 400

V2: 200

S: 75

Schedule:

8:00: school

12:00: playground

14:00: hospital

17:00: home

Agent - Based Model

Characteristic of Area B

States:

V1: 500

V2: 350

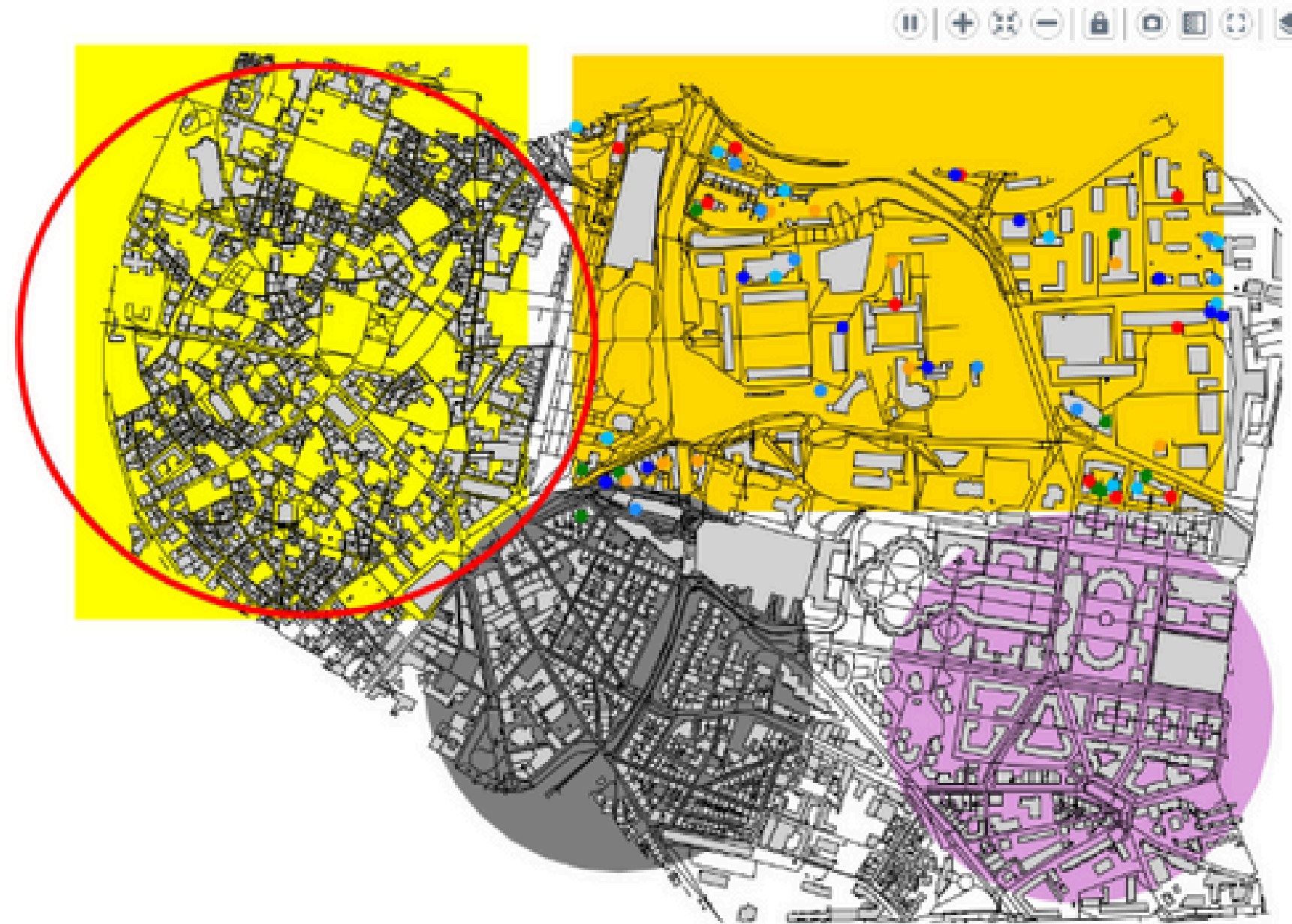
S: 50

Schedule:

9:00: school

15:00: hospital

16:00: home



Agent - Based Model

Characteristic of Area C



States

V1: 150

V2: 200

S: 10

Schedule:

10:00: school

13:00: playground

16:00: hospital

19:00: home

Agent - Based Model

Characteristic of Area D

States:

V1: 250

V2: 250

S: 20

Schedule:

11:00: school

14:00: playground

17:00: hospital

20:00 home



What is the difference?

Agent Based Models

- **Approach:** Agent
- **Simulation:** Individual
- **Heterogeneity:** Yes
- **Spatial analysis:** always
- **Interaction:** directly between individuals
- **Simulation:** vary scenarios

Compartment Models

- **Approach:** Compartment
- **Simulation:** Population
- **Heterogeneity:** No
- **Spatial analysis:** sometimes
- **Interaction:** between compartments
- **Simulations:** the same characteristic

REFERENCES



Lecture Guide
<https://bit.ly/4f9rVZe>



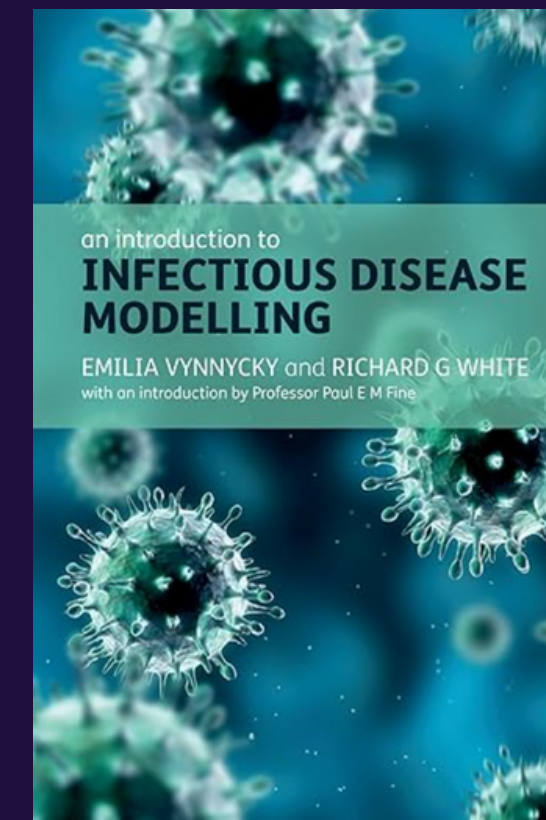
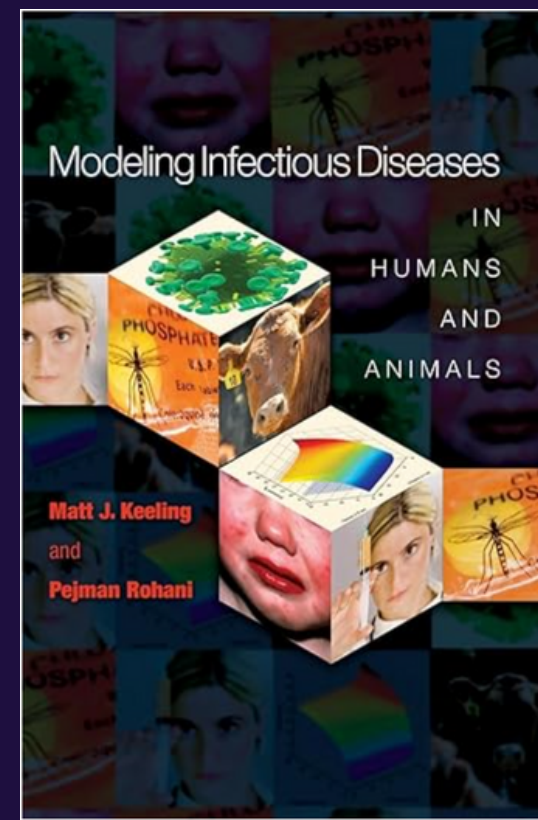
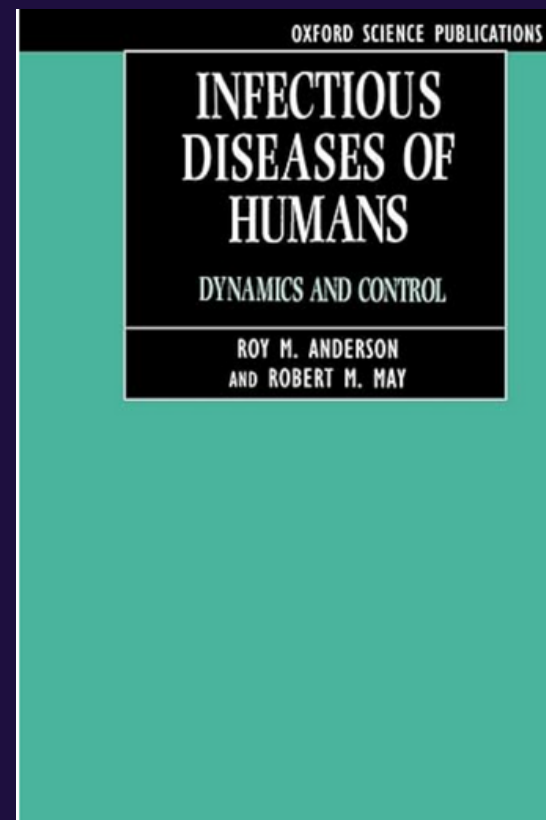
Modelling 102 Folder
<https://bit.ly/3Y8h8s5>



R code + Papers
<https://bit.ly/3Lq2Awu>



Dr. Thinh Ong
<https://drthinhong.com/idm4b/>





**THANK YOU
FOR LISTENING!**
QUESTIONS?